Introduction to Pharmacology & Drug Development

2026

Dr Grace Chua Pin Fen

MD, PhD

Unit Pharmacology

Department of Paraclinical Sciences

Welcome to Pharmacology

- Pharmacology Lecturers:

  A/P Dr Lawrence anak Anchah

  Dr Grace Chua Pin Fen

  Dr Dayang Erna Zulaikha binti Awang Hamsin

  Dr Muhammad Wahizul Haswan bin Abdul Aziz

  Dr Siti Zaleha binti Raduan

  Dr Melissa Lim Siaw Han

  Dr Melissa binti Mohammad Hirman

Pharmacology @ UNIMAS

Pharmacology Content in Block Two

- General Introduction to Pharmacology
- Dosage Forms & Routes of Administration
- Practical (Dose Response)
- Pharmacokinetics I-III
- Pharmacodynamics I-II
- *Physiology of Autonomic Nervous System &* Metabolism of Neurotransmitters
- Adrenergic agonists, antagonists
- Cholinergic agonists, antagonists
- Antimicrobials I-III
- Drug Adverse Effect & Toxicity

Learning Objectives

At the end of the teaching session, the student should be able to:

1. define pharmacology and contrast with the related disciplines of pharmacy and toxicology.
2. list the subdivisions of pharmacology.
3. state the key events in the development of modern pharmacology and drug use.
4. outline the key steps in the modern process of drug development.
5. explain the differences between the chemical, generic and proprietary names of drugs.
6. describe the content of the phase one pharmacology curriculum and reference resources.

Definition of Pharmacology

- Definition of a “drug” or “substance”
  - any compound that can **modify** the biological functions of living organisms

    -includes medicines, toxins, substance of abuse, caffeine/nicotine/alcohol, etc

Definition of Pharmacology

- Pharmacology
  - The study of substances/drug that interact with living systems through chemical processes
    - By binding to regulatory molecules→ activating/inhibiting normal body processes
    - In short, **the study of what drugs/substance do and how they do it.**

- Medical pharmacology
  - The study of pharmacological substances used to prevent, diagnose and treat disease.

Related Disciplines

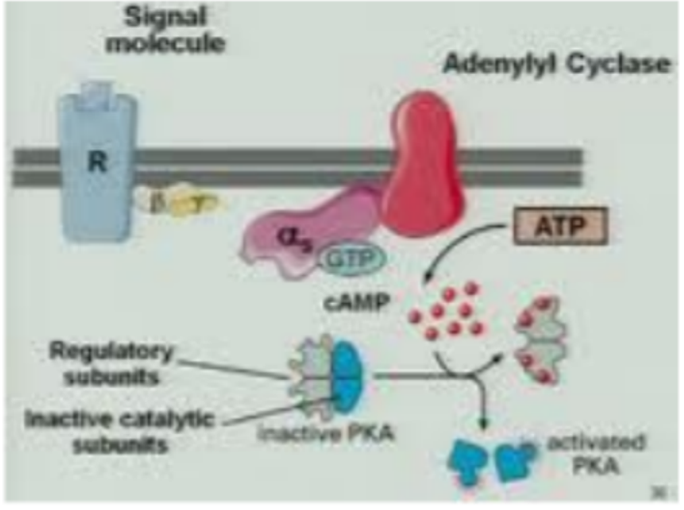
- Toxicology
  - A branch of pharmacology that deals with the undesirable effects on living systems.
  - The study of the harmful (adverse/toxic) effects of chemicals in biological systems .

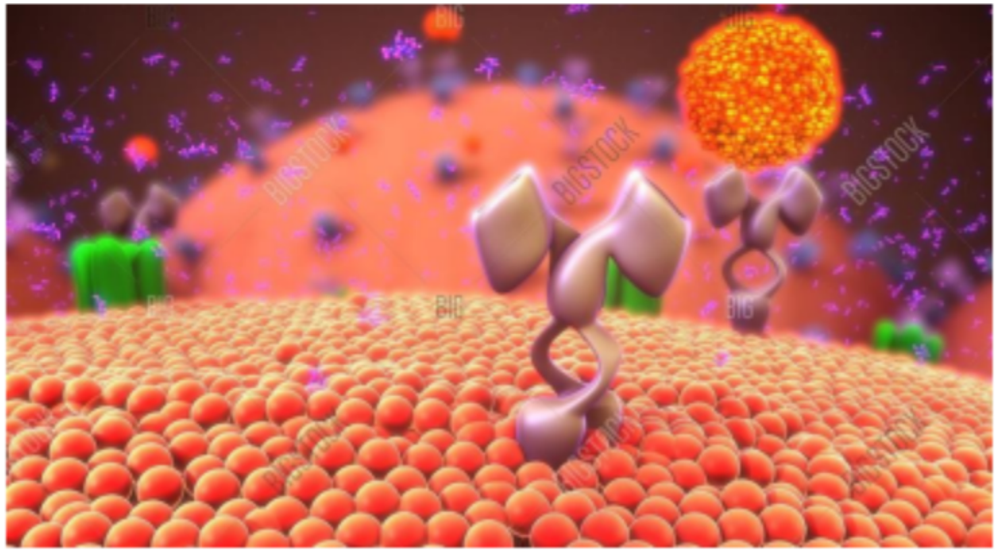
- Pharmacy
  - A profession focused on patient care and the safe, effective use of medications, including preparing, dispensing prescriptions and counseling patients.

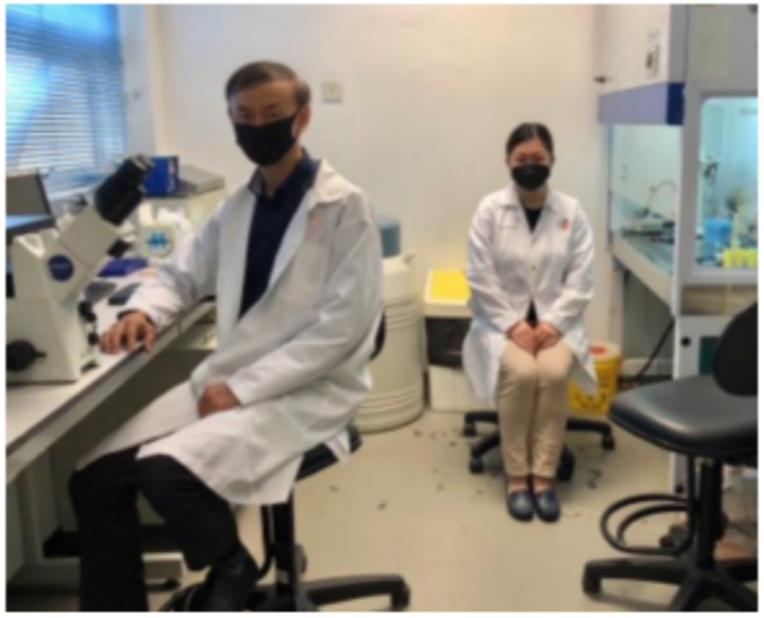
- Pharmacology is the science of drugs, while pharmacy is the practice of applying that science in patient care (E.g preparing, dispensing, and reviewing medication).

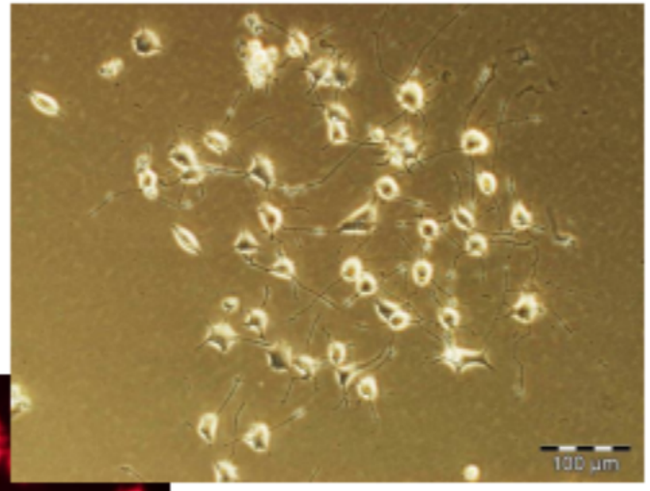
Divisions of Pharmacology

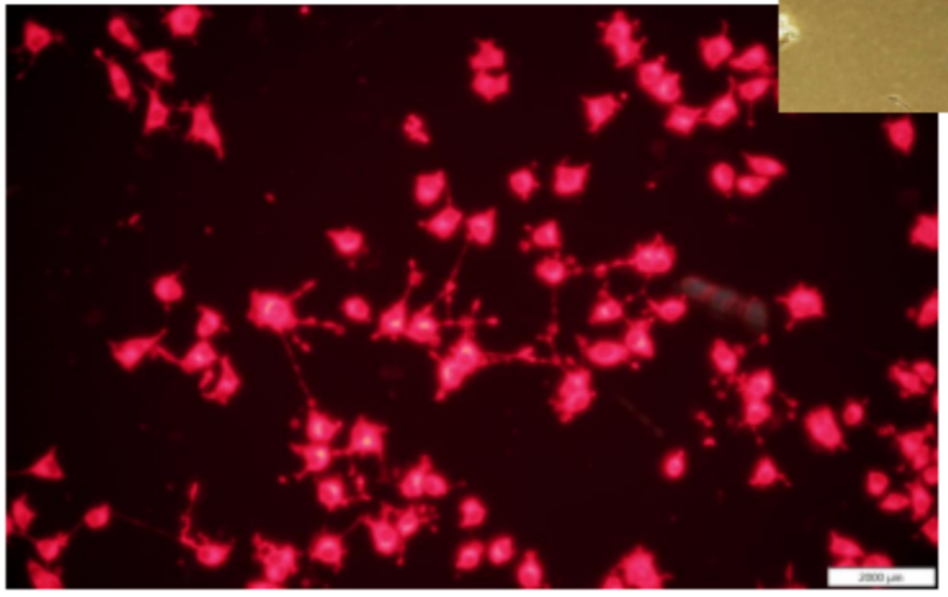
- Molecular Pharmacology:
  - The study of the cellular and molecular biology of drug action
    - understand molecular mechanism of drug action
    - identify molecular targets for drug intervention











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Optimisation of a PC12 cell-based in vitro stroke model for screening neuroprotective agents

PinFen Chua & William K. Lim

Contents lists available at ScienceDirect

Journal of King Saud University - Science

journal homepage: www.sciencedirect.com

Advancing Parkinson's disease biopathology and drug discovery by dual cellular modelling

PinFen Chua [a], Nurr Maria Ulfa Seruji [b], Mas Atikah Lizazman [b], Vivien Yi Mian Jong [b], William K. Lim [a,*]

[a] Department of Paraclinical Sciences, Faculty of Medicine and Health Sciences, Universiti Malaysia Sarawak, 94300 Kota Samarahan, Sarawak, Malaysia
[b] Centre of Applied Science Studies, Universiti Teknologi MARA Sarawak, 94300 Kota Samarahan, Sarawak, Malaysia

Division of Pharmacology

- Behavioural Pharmacology:
  - the study of how drugs affect behavior, memory, learning, addiction, etc

Division of Pharmacology

- Clinical Pharmacology:
  - the application of drugs for the prevention, treatment or diagnosis of human disease.

Division of Pharmacology

- Pharmacogenomic:
  - The study of genetic factors that underlie variation in drug response.

Pharmacologist training

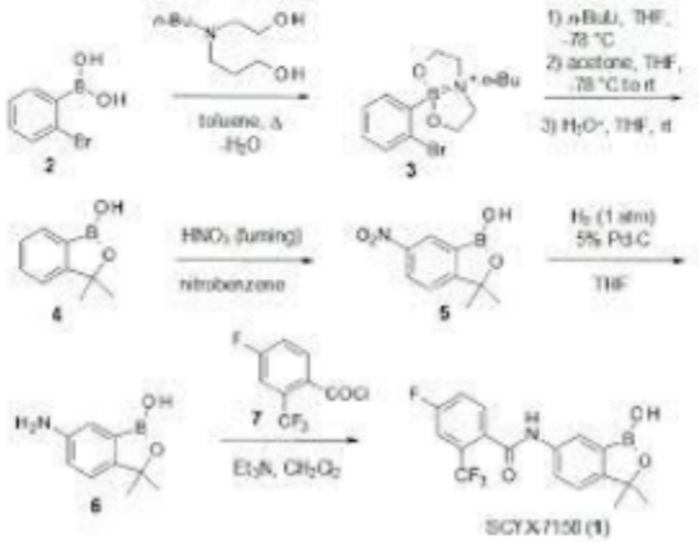
- Basic science degree or pharmacy degree → M.Sc/Ph.D in pharmacology
- Basic medical degree → specialist training in internal medicine (clinical pharmacology subspecialty)

Historical Development of Pharmacology: Ancient Times

- Ancient civilisations: disease is caused by gods, spirits, supernatural forces.
- Treatment with medicines accompanied by religious rituals

Historical Development of Pharmacology: Ancient to Modern

- Advances in chemistry and physiology lead to birth of pharmacology.

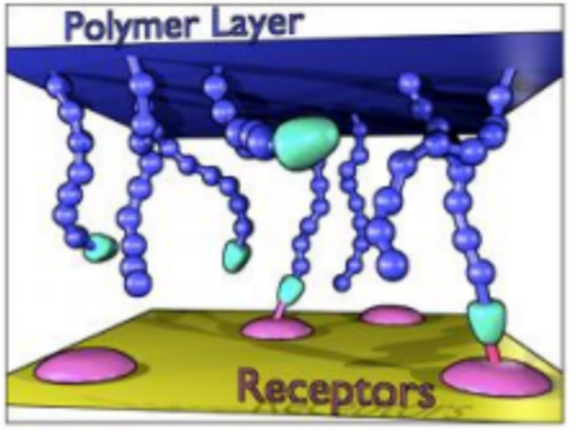


Historical Development of Pharmacology: Ancient to Modern

Beginning of modern pharmacology Rudolf Buchheim (1820-1879):

Drug activity is due to a reaction between the drug and cell components.

1st Institute for Pharmacology at University of Dorpat (Germany) in 1847



Historical Development of Pharmacology: Modern Era

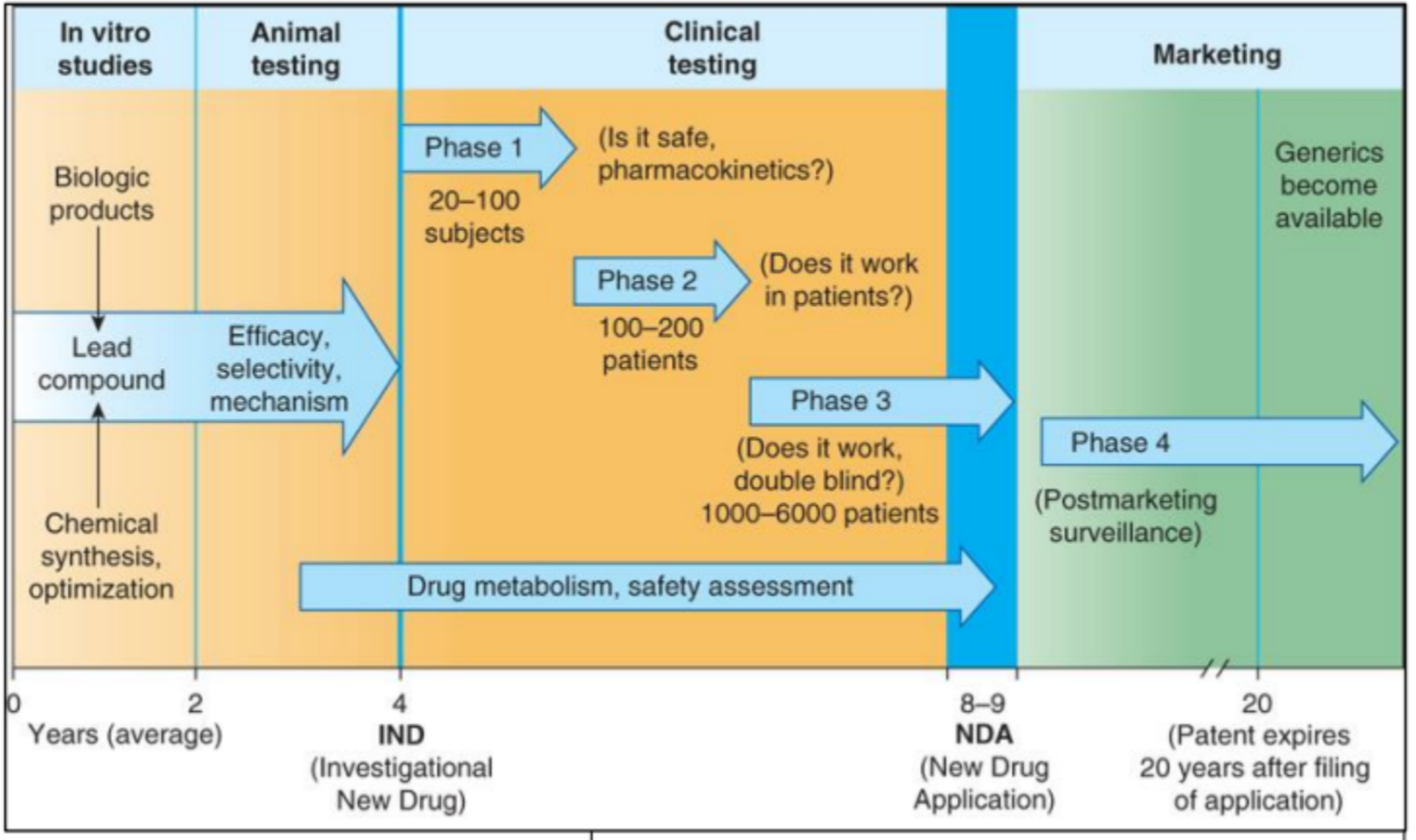
- **Early 20th century**: introduction of receptor theory; development of synthetic drugs
- **Mid-20th century**: discovery of antibiotics (penicillin, streptomycin); standardisation of clinical trials. penicillin by Alexander Fleming in 1928, which transformed medicine.
- **Late 20th century to present**: molecular pharmacology, receptor cloning, biotechnology drugs (monoclonal antibodies, recombinant proteins), rational drug design, and high-throughput screening.

Drug Discovery and Development

Key steps in the modern drug development process

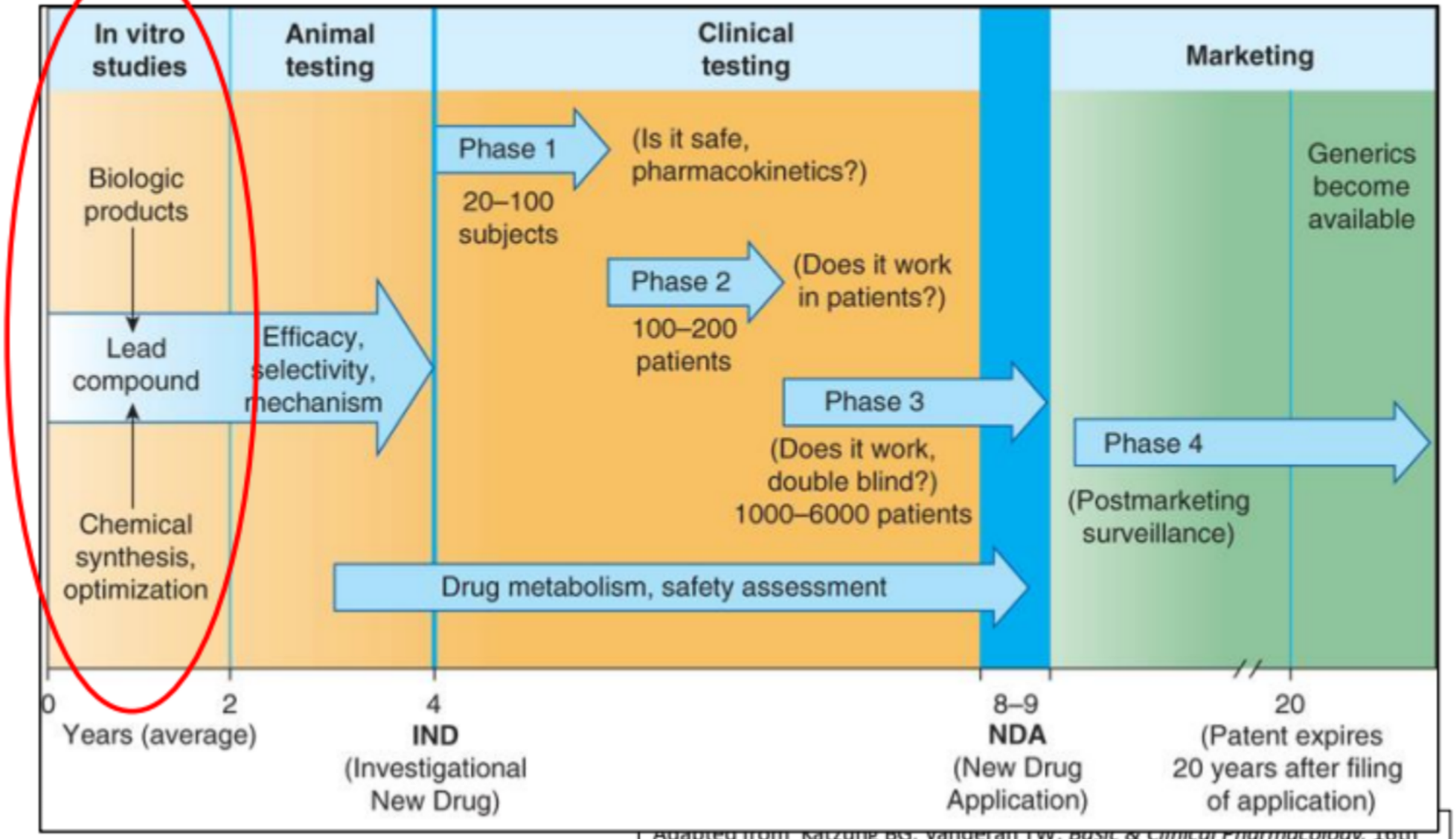
- **Drug discovery / target identification** – identify biological targets and potential lead compounds.
- **Preclinical testing** – laboratory and animal studies for efficacy, toxicity, pharmacokinetics.
- **Clinical trials**:
  - *Phase I*: safety, tolerability, pharmacokinetics in healthy volunteers.
  - *Phase II*: efficacy and dose-ranging in patients.
  - *Phase III*: large-scale trials for safety and effectiveness.
- **Regulatory approval** – submission to authorities (e.g., FDA (Food and Drug Administration), EMA (European Medicines Agency)).
- **Post-marketing surveillance (Phase IV)** – monitoring long-term effects, rare adverse events, real-world effectiveness.

Drug Discovery and Development



Adapted from Katzung BG, Vanderah TW: *Basic & Clinical Pharmacology,* 16th ed. New York,

Drug Discovery and Development



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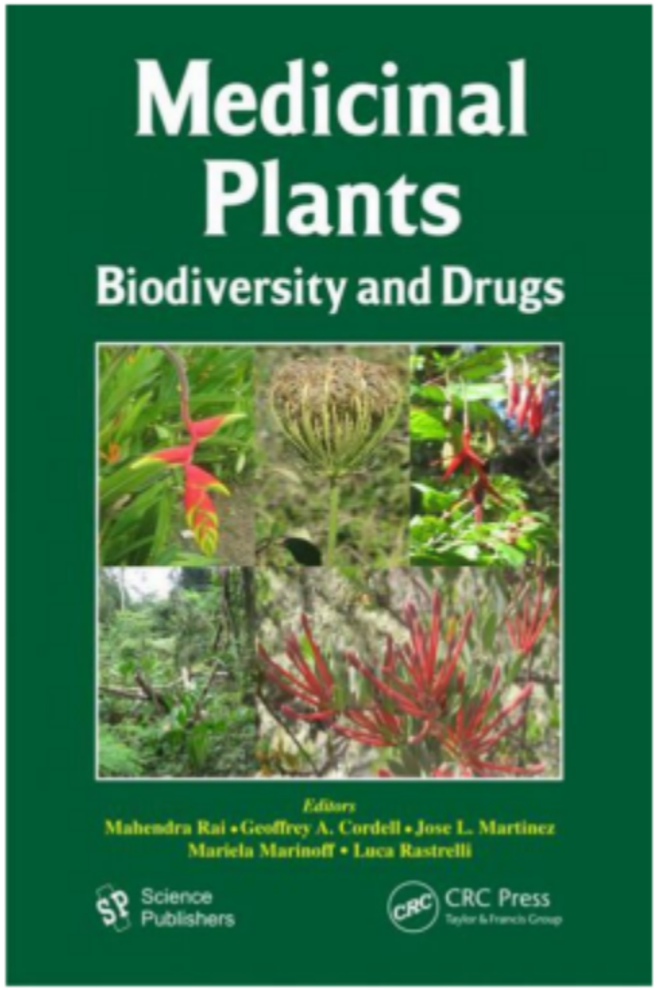
Drug Discovery and Development

New drugs are derived from

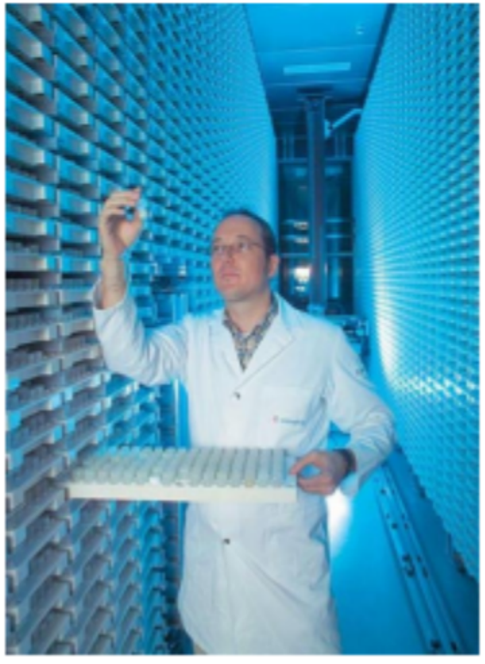
1. Natural product

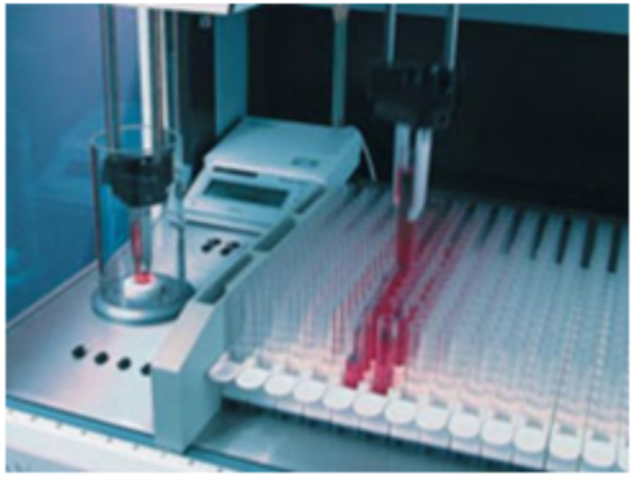
E.g active ingredients from plants



Drug Discovery and Development

- New drugs are derived from:
  2. Screening chemical library for activity.

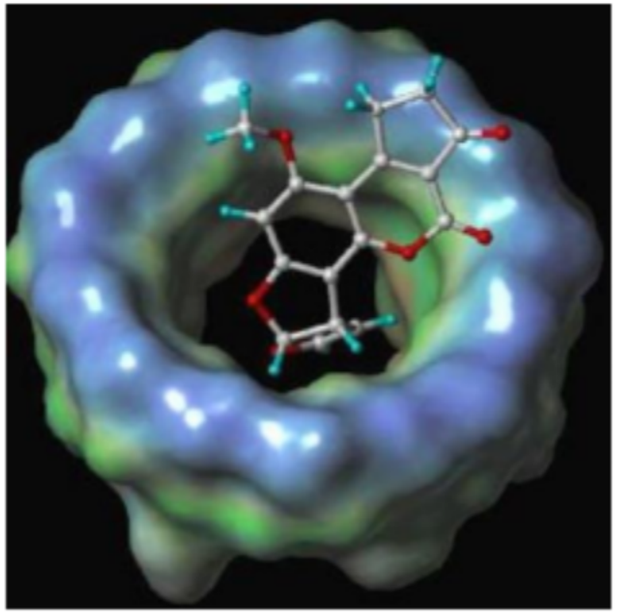




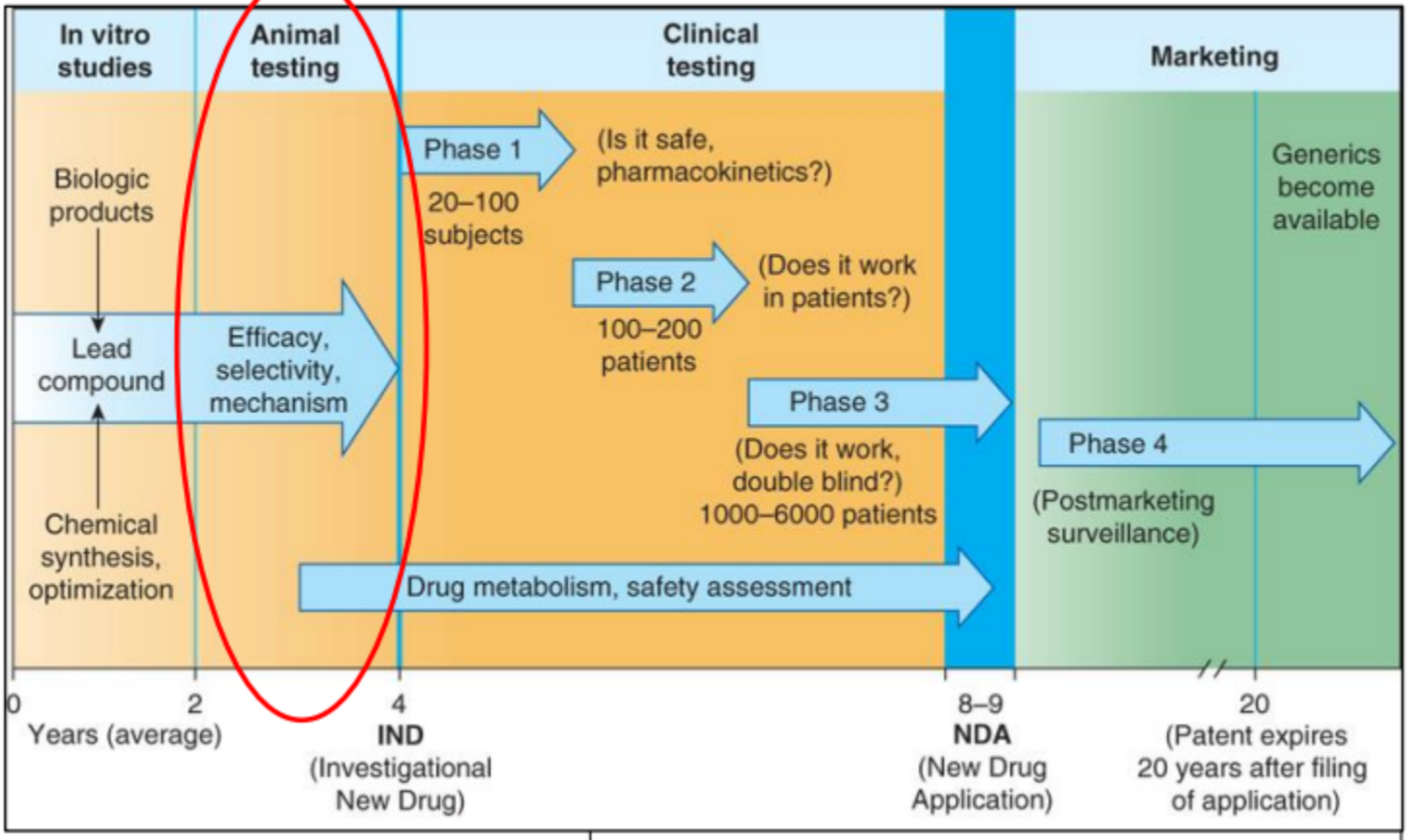
Drug Discovery and Development

- New drugs are derived from:

  3. synthesis of compounds based on computer modeling and molecular biology/chemistry research. E.g *computer-aided drug discovery* (CADD)



Drug Discovery and Development



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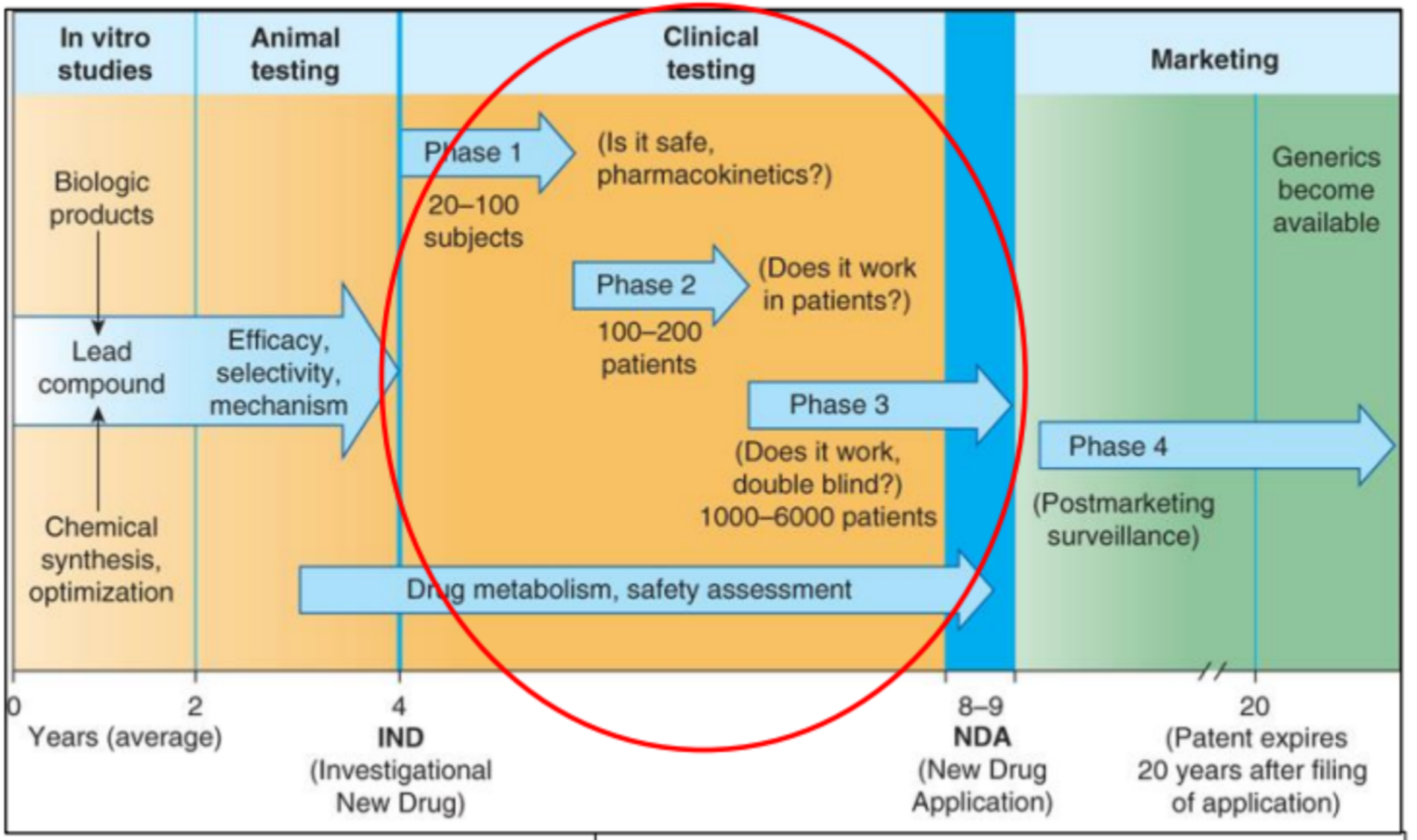
Animal testing

- laboratory and animal studies for efficacy, toxicity, pharmacokinetic.
- Goals are to
  - identifying potential human toxicities.
  - designing tests to further define the toxic mechanisms.
  - predicting the most relevant toxicities to be monitored in clinical trials.

Drug Discovery and Development



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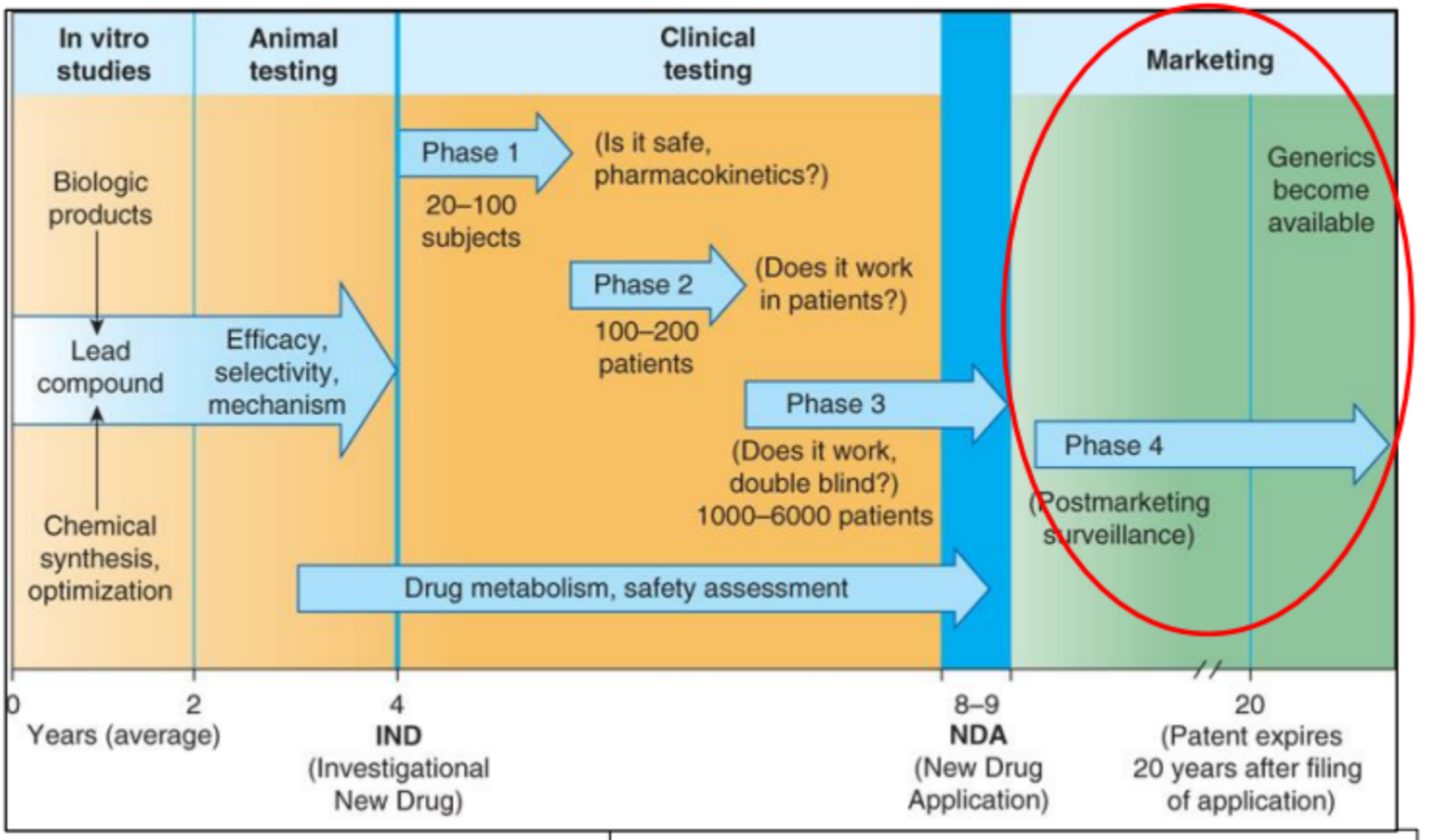
Clinical testing – Evaluation in Humans

- Very small number of lead compounds reach clinical trials
- Less than 1/3 of the drugs survive clinical trials and reach marketplace.
- Requires at least 4- 6 years
- Divided into 3 phases
  - Phase I
  - Phase II
  - Phase III

Drug Discovery and Development

- Phase I
  - safety, tolerability, pharmacokinetics in healthy volunteers.
- Phase II
  - efficacy and dose-ranging in patients.
- Phase III
  - large-scale trials for safety and efficacy.
  - the drug is evaluated in much larger numbers of patients—usually thousands—of patients with the target disease to further establish and confirm safety and efficacy.

Drug Discovery and Development



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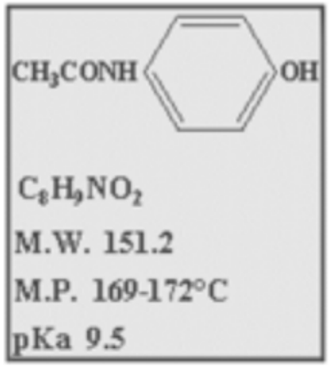
Drug Discovery and Development

Post-marketing surveillance (Phase IV) – monitoring long-term effects, rare adverse events, real-world effectiveness.

After patent expired, generic drug will enter the market.

Naming of Drugs



Differences between chemical, generic, and proprietary names of drugs

1. Chemical name: precise description of the molecular structure (e.g., *N-acetyl-p-aminophenol*).

2. Generic (non-proprietary) name: internationally accepted standard name assigned by bodies like WHO (e.g., *paracetamol (US name: acetaminophen)*).

3. Proprietary (brand / trade) name: name chosen by a company for marketing (e.g., *Panadol®, Tylenol®*).
→ One drug can have many brand names but only one generic name.

Naming of Drugs

- Generic name is used in teaching, exam and prescription.

- Tip: Sometimes common ending of generic names indicates pharmacological class of drugs:

  e.g. β adrenergic receptor blockers:

Propranolol, metoprolol, atenolol, oxprenolol

  e.g. Angiotensin Converting Enzyme (ACE) Inhibitors

Captopril, enalapril, lisinopril, ramipril, quinapril

- not for Nonsteroidal Antiinflammatory Drugs (NSAID)

Aspirin, Ibuprofen, Diclofenac, Mefenamic Acid

Reference Books

1. Katzung, B.G., Trevo,r A.J. Basic and Basic and Clinical Pharmacology (Latest edition). McGraw-Hill Education / Medical
2. Brunton, L.L., Chabner, B.A., Knollman, B.C. Goodman and Gilman's The Pharmacological Basis of Therapeutics (latest edition). McGraw-Hill Education / Medical
3. Shargel, L., Yu, A.B.C. Applied biopharmaceutics and pharmacokinetics (Latest edition). McGraw-Hill Education / Medical
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8. Whalen, K. (Ed.). *Lippincott's Illustrated Reviews: Pharmacology.* Wolters Kluwer.

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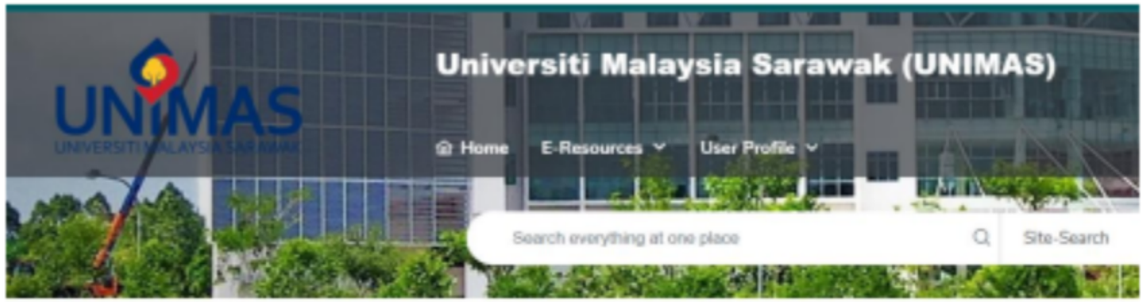

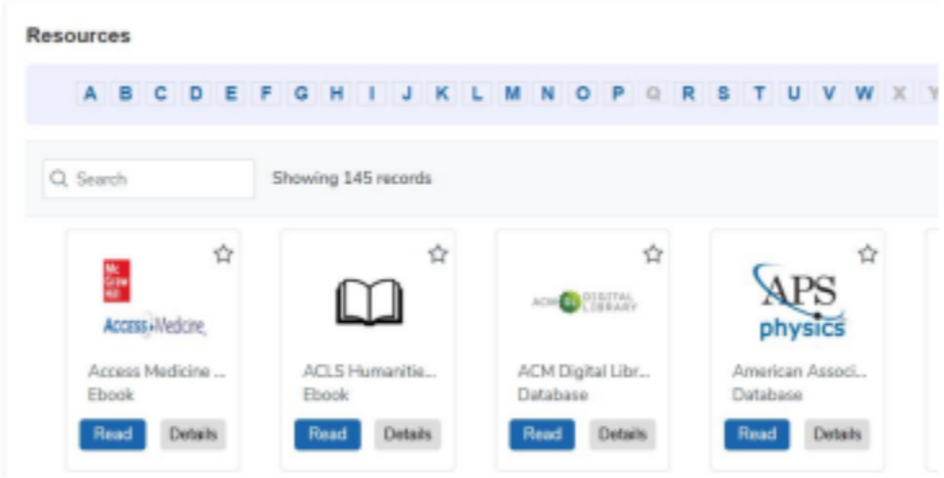

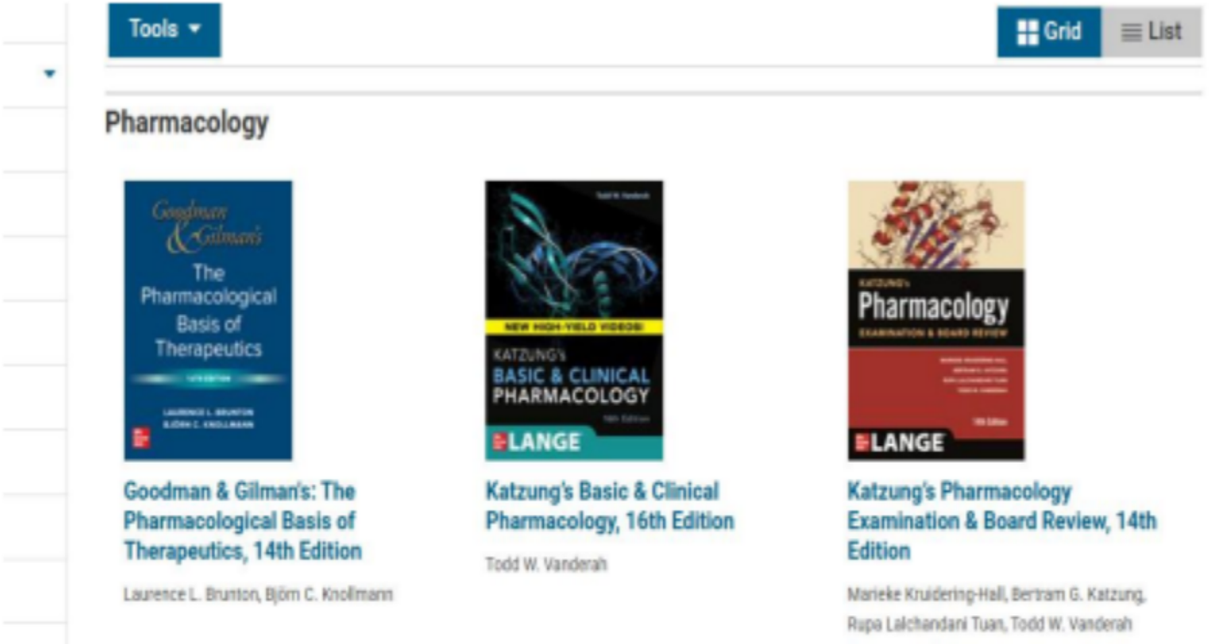

Pharmacology Curriculum

- Block 2 pharmacology content:
  - General Introduction to Pharmacology
  - Dosage Forms & Routes of Administration
  - Pharmacokinetics I-III
  - Pharmacodynamics I-II
  - Physiology of Autonomic Nervous System & Metabolism of Neurotransmitters
  - Adrenergic agonists, antagonists
  - Cholinergic agonists, antagonists
  - Antimicrobials I-III
  - Drug Adverse Effect & Toxicity

Pharmacology Curriculum

- Blocks 3 - 10:

PBL or lectures covering important drugs found in that body system.

Thank you

- Email- fcpin@unimas.my